

Archie Tansaria

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EDUCATION

Nitte Meenakshi Institute of Technology | Bengaluru

2023 - 2027

Bachelors in Engineering in Electronics and Communications Engineering | **GPA: 8.08/10.0** (current)

WORK EXPERIENCE

ASE Club, Software Developer Intern | Bengaluru

June 2024 - Nov 2025

- Developed and maintained 3 production-grade projects with a focus on modern, scalable fullstack development.
- Designed and launched a campus-wide resource sharing platform enabling 500+ students to lend/borrow items like books, lab tools and electronics eliminating traditional paper-based tracking.

TECHNICAL PROJECTS

[DURA] AI-native Dependency Update Risk Analysis Engine | TypeScript, Node.js, React, AWS (ECS Fargate, Lambda, Elasticache), Redis, BullMQ, Cloudflare, MongoDB, Docker, MCP [Github](#) [Live](#)

- Architected a distributed dependency risk analysis engine as a GitHub App, enabling real-time PR risk evaluation via a React dashboard and secure webhook ingestion using a Cloudflare Tunnel sidecar.
- Designed a **cost-optimized event-driven system** using AWS Lambda to trigger ECS Fargate workers from zero to one on-demand, reducing idle infrastructure costs by ~70%.
- Built an **asynchronous job processing pipeline** using BullMQ & Redis with containerized workers and stalled-job recovery to ensure 100% task completion within serverless constraints.
- Developed a risk-scoring engine using semantic version analysis and release-note parsing, optimized with parallel metadata fetching and Redis caching.
- Created a Node.js **CLI**, **MCP server** and **Apify Actor** enabling AI-agent integration and local repository analysis.

[Loom] Psychology-Based Compatibility Matching Platform | React (Vite), Node.js, Express, MongoDB, Socket.IO, JWT, Docker [Github](#) [Live](#)

- Developed full-stack dating application with React.js (Vite) SPA, Express.js middleware pipeline, MongoDB with Mongoose schema validation, and Socket.IO event-driven real-time messaging infrastructure.
- Constructed **multi-dimensional compatibility algorithm** with weighted trait analysis and compound-indexed MongoDB aggregation pipelines (sub-200ms queries).
- Built production-grade security layer: JWT refresh token rotation, bcrypt adaptive hashing, CORS policies, express-validator sanitization, and Socket.IO namespace isolation for chat scalability.

SKILLS

Languages: JavaScript, TypeScript, Python, C++, C; **Frontend:** React, Next.js, Redux, Three.js; **Testing:** Jest

Backend: Node.js, Express.js, Socket.IO, Prisma; **Databases:** MongoDB, MySQL, PostgreSQL, Redis

DevOps/Cloud: Firebase, Docker, Vercel, Render, AWS, GitHub Actions; **Tools:** Figma, MATLAB, TMUX, Neovim

AWARDS AND HONOURS

- **Winner** – HackThisFall **DevDay Hackathon** (2025), built DURA Apify Actor (Dependency Risk Automation)
- **Winner** – **HackToFuture Hackathon** (2025), built a Decentralized Patient Health Record System
- **1st Runner Up** – **I-Solve Hacks** (2024), developed a Corporate Carpooling Optimization Platform
- **"Best Use of Data" Award** – **MLH Hack Week** (2023)